

The treewidth and pathwidth of hypercubes[☆]

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Abstract

The d -dimensional hypercube, H_d , is the graph on 2^d vertices, which correspond to the 2^d d -vectors whose components are either 0 or 1, two of the vertices being adjacent when they differ in just one coordinate. The notion of Hamming graphs (denoted by K_q^d) generalizes the notion of hypercubes: The vertices correspond to the q^d d -vectors where the components are from the set $\{0, 1, 2, \dots, q-1\}$, and two of the vertices are adjacent if and only if the corresponding vectors differ in exactly one component. In this paper we show that the $\text{pw}(H_d) = \sum_{m=0}^{d-1} \binom{m}{2}$ and the $\text{tw}(K_q^d) = \theta(q^d/\sqrt{d})$.

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1. Introduction

The notions of treewidth and pathwidth (see definitions below) were introduced by Robertson and Seymour [9,8] in their series of fundamental papers on graph minors. In this paper we estimate the values of these parameters for hypercubes and Hamming graphs.

The d -dimensional hypercube, H_d , is the graph on 2^d vertices, which correspond to the 2^d d -vectors whose components are either 0 or 1, two of the vertices being adjacent when they differ in just one coordinate. Hypercubes are a well-studied class of graphs, which arise in the context of parallel computing, coding theory, algebraic graph theory and many other areas. They are popular because of their symmetry, small diameter and many interesting graph-theoretic properties.

Our first result in this paper is about the pathwidth of hypercubes. We show that $\text{pw}(H_d) = \sum_{m=0}^{d-1} \binom{m}{2}$. Moreover, we exhibit an optimal path decomposition of H_d . For treewidth, we consider a more general setting: The Hamming graph K_q^d is the graph on q^d vertices which correspond to the q^d d -vectors, with coordinates coming from a set of size q (say $\{0, 1, 2, \dots, q-1\}$), two of the vertices being adjacent iff the corresponding vectors differ in exactly one coordinate. (Clearly, the d -dimensional hypercube H_d is a special case of the Hamming graphs, K_q^d , namely when $q = 2$.) We show that the treewidth of K_q^d is $\theta(q^d/\sqrt{d})$.

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Definition 1. A tree decomposition of $G = (V, E)$ is a pair (X, T) where $X = \{X_i : i \in I\}$ is a collection of subsets of V (we call these subsets the nodes of the decomposition) and $T = (I, F)$ is a tree having the index set I as the set of vertices such that the following conditions are satisfied:

1. $\bigcup_{i \in I} X_i = V$.
2. $\forall (u, v) \in E, \exists i \in I : u, v \in X_i$.
3. $\forall i, j, k \in I$: if j is on a path in T from i to k , then $X_i \cap X_k \subseteq X_j$.

The width of a tree decomposition $(\{X_i : i \in I\}, T)$ is $\max_{i \in I} |X_i| - 1$.

The treewidth of G is the minimum width over all tree decompositions of G and is denoted by $\text{tw}(G)$.

A path decomposition of $G = (V, E)$ is a tree decomposition (X, T) in which T is required to be a path. The pathwidth of G is defined to be the minimum width over all path decompositions of G and is denoted by $\text{pw}(G)$.

All the results we present in this paper are consequences of the results on the vertex isoperimetric property of the hypercube and Hamming graphs, discovered by Harper [5,6]. Let $G(V, E)$ be a graph. For $S \subseteq V$, the vertex boundary $N(S)$ can be defined as follows:

$$N(S) = \{w \in V - S : \exists v \in S \text{ such that } \{w, v\} \in E\}. \quad (1)$$

The vertex isoperimetric problem is to minimize $|N(S)|$ over all subsets S of V with $|S| = \ell$ for a given integer ℓ . We denote this minimum value by $b_v(\ell, G)$, i.e., $b_v(\ell, G) = \min_{S \subseteq V, |S|=\ell} |N(S)|$. Harper studied the vertex isoperimetric problem of the hypercubes (and later, that of the Hamming graphs) in order to estimate the value of a parameter called the *bandwidth*.

The bandwidth of a graph $G(V, E)$ can be defined as follows: A bijection $\phi : V \rightarrow \{1, \dots, n\}$ is called an ordering of the vertices of G . Then for any edge $e = \{u, v\} \in E$, let $\Delta(e, \phi) = |\phi(u) - \phi(v)|$. The bandwidth of G is the minimum over all possible orderings ϕ of $V(G)$ of the maximum value of $\Delta(e, \phi)$ over all edges $e \in E$.

It is rather easy to observe from the definitions of treewidth, pathwidth and bandwidth, that the following inequality exists between these parameters. (See [1] for a proof.)

$$\text{treewidth} \leq \text{pathwidth} \leq \text{bandwidth}. \quad (2)$$

In general the inequalities given above are strict. Moreover, for any pair of these parameters, there exist classes of graphs, for which the gap between the values of those parameters grows as the number of vertices increases.

In this paper we show that, using isoperimetric properties one can in fact give lower bounds for treewidth and pathwidth also, not just for bandwidth. (Harper's original intention of studying the isoperimetric problem was to lower bound the bandwidth.) Thus it turns out that in the case of hypercubes, bandwidth equals pathwidth. This is indeed one more addition to the list of cute graph theoretic properties, which the hypercubes possess. Currently we are only able to estimate the value of the treewidth up to a constant factor. But it is possible that the treewidth itself is equal to the pathwidth and bandwidth, for H_d .

2. Pathwidth of the hypercube

A graph $G(V, E)$ is defined to be an interval graph iff its vertices $V = \{v_1, v_2, \dots, v_n\}$ can be put in one to one correspondence with a set of intervals $\{I_1, I_2, \dots, I_n\}$ on the real line in such a way that $\{v_i, v_j\} \in E$ if and only if the corresponding intervals I_i and I_j have a non-empty intersection. Without loss of generality one can assume that all the intervals are closed intervals, see for example [4, p. 13].

The clique number $\omega(G)$ is defined to be the number of vertices in a maximum sized clique in G . The notion of pathwidth is related to the interval completion of a graph, in the following way. (See [1] for a proof of Lemma 1 and for a discussion of related ideas.)

Definition 2. A graph G has interval-width, $\text{IW}(G) = k$ iff k is the smallest non-negative integer such that G is a subgraph of some interval graph H , with $\omega(H) = k + 1$.

Lemma 1. Let G be a graph. Then $\text{pw}(G) = \text{IW}(G)$.

First we develop a lower bound for the pathwidth of a graph in terms of the vertex isoperimetric property.

Lemma 2. Let $G(V, E)$ be a connected interval graph and $1 \leq s \leq n$. Then $\omega(G) \geq b_v(s, G) + 1$.

Proof. Consider an interval representation of G , with closed intervals on the real line. Let $\{I_1, I_2, \dots, I_n\}$ be the set of intervals, ordered from 1 to n in the non-decreasing order of their right end points. Let v_i be the vertex corresponding to I_i , $1 \leq i \leq n$. Let $X = \{v_1, v_2, \dots, v_s\}$. Let P be the right end point of I_s . Each interval I , corresponding to a vertex v in $N(X)$ contains P : the right end point of I must be to the right of or equal to P as $v \notin X$, and since I must intersect with an interval in X , its left end point must be to the left of or equal to P . Now, $N(X) \cup \{v_s\}$ induces a clique as each pair of intervals corresponding to the vertices in $N(X) \cup \{v_s\}$ have P in their intersection. Thus, $\omega(G) \geq |N(X)| + 1 \geq b_v(s, G) + 1$. $\square$

Theorem 1. Let $G(V, E)$ be any graph on n vertices, and let $1 \leq s \leq n$. Then $\text{pw}(G) \geq b_v(s, G)$.

Proof. Clearly in any interval super graph G' of G (on the same set of vertices) $b_v(s, G') \geq b_v(s, G)$. The theorem follows from Lemmas 1 and 2. $\square$

Harper proved the following result about the bandwidth of H_d .

Lemma 3. $\text{bandwidth}(H_d) = \max_s b_v(s, H_d) = \sum_{m=0}^{d-1} \binom{\frac{m}{2}}{\frac{m}{2}}$ (The maximum is taken over s in the range $1 \leq s \leq 2^d$).

The last equality can be proved by induction and is mentioned in the last page of Harper's article [5]. Putting together all the pieces, we get our result on the pathwidth of H_d .

Theorem 2. $\text{pw}(H_d) = \text{bandwidth}(H_d) = \max_s b_v(s, H_d) = \sum_{m=0}^{d-1} \binom{\frac{m}{2}}{\frac{m}{2}}$.

Proof. By Theorem 1, we know that $\text{pw}(H_d) \geq \max_s b_v(s, H_d)$. It follows by Lemma 3, that $\text{pw}(H_d) \geq \text{bandwidth}(H_d)$. But we know that bandwidth of any graph is at least as much as its pathwidth (inequality (2)). The required result follows. $\square$

Remark. By Sterling's approximation, it can be shown that $\sum_{m=0}^{d-1} \binom{\frac{m}{2}}{\frac{m}{2}} = O\left(\frac{2^d}{\sqrt{d}}\right)$.

3. An optimal path decomposition for H_d

In this section, we will exhibit a path decomposition of H_d , which is optimum. In fact, this path decomposition corresponds to Harper's optimal layout for bandwidth. (The reader may note that just as pathwidth of G can be expressed in terms of the clique number of an optimum interval super graph of G (see Lemma 1), bandwidth of G can be expressed in terms of the clique number of an optimum proper interval super graph of G . One can construct an interval super graph of H_d corresponding to the path decomposition of this section in a natural way, and this interval graph is the same as the proper interval super graph which can be constructed from Harper's optimum layout for bandwidth in a similar way. In this sense, the construction here is only a restatement of Harper's layout which achieves optimum bandwidth for H_d , to suit the definition of path decomposition.)

We need to review a few concepts from the theory of set systems. (For a comprehensive exposition of these concepts, see [2, Chapter 5].)

The Colex ordering: Let $X = \{1, 2, \dots, d\}$. Let $X^{(r)} = \{S \subseteq X : |S| = r\}$. The Colex order on $X^{(r)}$, the set of all r -subsets of X , is defined as follows: Let $A = \{a_1, a_2, \dots, a_r\}$ and $B = \{b_1, b_2, \dots, b_r\}$ be two different r -subsets of X . (We assume that $a_1 \leq a_2 \leq \dots \leq a_r$ and $b_1 \leq b_2 \leq \dots \leq b_r$.) According to Colex order, $A < B$ if $A \neq B$ and for $s = \max\{t : a_t \neq b_t\}$ we have $a_s < b_s$. In other words, $A < B$, in the colex order if either $a_r < b_r$ or $a_r = b_r$ and

$a_{r-1} < b_{r-1}$ or $a_r = b_r$ and $a_{r-1} = b_{r-1}$ and $a_{r-2} < b_{r-2}$ or $\cdots$ or $a_r = b_r$ and $a_{r-1} = b_{r-1}$ and $a_{r-2} = b_{r-2} \cdots a_2 = b_2$ and $a_1 < b_1$.

The lower shadow of a set system: Let $\mathcal{A} \subseteq X^{(r)}$ be a collection of r -subsets of X . The lower shadow of $\mathcal{A}$, $\partial_\ell^{(r)}(\mathcal{A})$ is defined as

$$\partial_\ell^{(r)}(\mathcal{A}) = \{B \in X^{(r-1)} : B \subseteq A, \text{ for some } A \in \mathcal{A}\}.$$

If $r = 0$, and $\emptyset \neq \mathcal{A} \subseteq X^{(0)} = X^{(0)} = \{\emptyset\}$, then $\mathcal{A} = \{\emptyset\}$ and $\partial_\ell^{(0)}(\mathcal{A}) = \emptyset$.

Note that the vertices of H_d can be put in one to one correspondence with the subsets of $X = \{1, 2, \dots, d\}$. For example, vector $(0, 0, \dots, 0)$ corresponds to the empty set and $(1, 1, \dots, 1)$ corresponds to $X = \{1, 2, \dots, d\}$. The subset S_v corresponding to a given vector v is determined as follows: $i \in S_v$ if and only if the i th component of the vector v is 1. We number the vertices of H_d , in the following way.

1. The vertex which corresponds to the set X which is the unique element of $X^{(d)}$ will be numbered 1 and will be denoted as v_1 .
2. If a vertex v corresponds to a subset in $X^{(r)}$ (where $0 \leq r \leq d-1$) and that subset is the m th subset in $X^{(r)}$ in the colex order, then v will be numbered $\sum_{i=r+1}^d \binom{d}{i} + m$. (Thus if $r_1 < r_2$, a vertex corresponding to a subset in $X^{(r_1)}$ gets a *higher number* than the vertex which corresponds to a subset in $X^{(r_2)}$. Within $X^{(r)}$ for a fixed r , the vertices get numbers according to the colex order of the corresponding subsets.) Finally, if a vertex is numbered i , it will be denoted as v_i .

Remark. If i is an integer, where $2 \leq i \leq 2^d$, it is easy to see that there is a unique pair (r, m) such that

$$i = \sum_{i=r+1}^d \binom{d}{i} + m,$$

where $0 \leq r \leq d-1$ and $0 < m \leq \binom{d}{r}$. Thus, the numbering scheme described above numbers the vertices of H_d from 1 to 2^d . For the ease of presentation, we will use the notation (r, m) to represent an integer i , if $i = \sum_{i=r+1}^d \binom{d}{i} + m$. We say that i is of the form (r, m) . Note that, when i is of the form (r, m) , the vertex v_i corresponds to a subset in $X^{(r)}$ and this subset will be the m th member in the colex order of $X^{(r)}$. To represent $i = 1$, we will use $(d, 1)$.

A path decomposition of H_d : Corresponding to each i , $i = 1$ to 2^d , we define a subset P_i as follows. Suppose i is of the form (r, m) . Then, let A_i be the set of first m members of $X^{(r)}$ in the colex order. (Note that A_i contains the subset S_{v_i} which corresponds to v_i .) Let $B_i = X^{(r)} - A_i \cup \{S_{v_i}\}$. We define P_i as,

$$P_i = B_i \cup \partial_\ell^{(r)}(A_i).$$

First we show that the sequence $(P_1, P_2, \dots, P_{2^d})$ of subsets (of vertices of H_d), defined above is indeed a path decomposition of H_d . We need to verify that the three conditions of Definition 1 are satisfied.

1. For every i , $1 \leq i \leq 2^d$, $v_i \in P_i$. Thus condition 1 is trivially satisfied.
2. Let (v_i, v_j) be an arbitrary edge in H_d . Without loss of generality, let $i < j$ and let v_i correspond to a subset S_{v_i} in $X^{(r)}$, for $1 \leq r \leq d$. Then clearly v_j corresponds to a subset S_{v_j} in $X^{(r-1)}$. Since $S_{v_i} \in A_i$, clearly $S_{v_j} \in \partial_\ell^{(r)}(A_i)$. It follows that $P_i = B_i \cup \partial_\ell^{(r)}(A_i)$ contains both v_i and v_j . Thus condition 2 is also satisfied.
3. Finally, we have to show that if a vertex x is in P_j and in P_k (where $j < k$), then $x \in P_i$ for all i , $j \leq i \leq k$. Consider an arbitrary vertex $x = v_t$ where $t = (r, m)$ for some r and m . By the definition of the sets P_i , any set P_i which contains v_t is such that $i \leq t$. Moreover every set P_i where i is of the form (r, m_0) where $m_0 \leq m$ contains v_t since for such i , obviously $B_i \supseteq B_t$. Also, let the lowest value of i , such that $v_t \in P_i$ be k . Clearly k will be of the form $(r+1, m_1)$ (except of course for $t = 1$, in which case v_t is only in P_1). Now, from the definition of P_i every set

P_i where i is of the form $(r+1, m_2)$ where $m_2 \geq m_1$ contains v_i since for any such i , $A_i \supseteq A_k$ and thus clearly $\partial_\ell^{(r+1)}(A_i) \supseteq \partial_\ell^{(r+1)}(A_k)$. Thus condition 3 is also satisfied.

Let m be an integer such that $0 < m \leq \binom{d}{r}$. Let $\mathcal{A}$ be the set of first m r -subsets in $X^{(r)}$ in the colex order. Then we use the symbol $\partial_\ell^{(r)}(m)$ to denote the cardinality of the lower shadow of $\mathcal{A}$. That is,

$$\partial_\ell^{(r)}(m) = |\partial_\ell^{(r)}(\mathcal{A})|.$$

The famous Kruskal–Katona theorem states that for any $\mathcal{A} \subseteq X^{(r)}$ where $|\mathcal{A}| = m$, $|\partial_\ell^{(r)}(\mathcal{A})| \geq \partial_\ell^{(r)}(m)$. Also it can be shown that if $m = \sum_{i=s}^r \binom{m_i}{i}$ ($1 \leq s \leq m_s < m_{s+1} < \dots < m_r$) then $\partial_\ell^{(r)}(m) = \sum_{i=s}^r \binom{m_i}{i-1}$ (see [2, Chapter 3]).

The following result is originally due to Harper [5]. Different proofs were given by [7,3]. We are following the exposition due to Bollobas [2, Chapter 16, p. 129].

Lemma 4. Let i , where $1 \leq i \leq 2^d$, be an integer of the form (r, m) . Then $b_v(i, H_d) = \binom{d}{r} - m + \partial_\ell^{(r)}(m)$.

Lemma 5. The path decomposition described above has the optimum width.

Proof. Clearly, the width of the above path decomposition is given by $\max_i |P_i| - 1$. But $|P_i| = |B_i| + |\partial_\ell^{(r)}(A_i)| = \binom{d}{r} - (m-1) + \partial_\ell^{(r)}(m)$, where i is of the form (r, m) . Thus in view of Lemma 4, the width of the path decomposition given above is the same as $\max_i b_v(i, H_d)$. But by Theorem 2, this is indeed the pathwidth of H_d . $\square$

4. Treewidth of Hamming graphs

The following Lemma is proved by Robertson and Seymour [8].

Lemma 6. Let $G(V, E)$ be a graph on n vertices and $Q \subseteq V$. Then if there exists a tree decomposition of G with width $< k$, then there exists a subset X of V with $|X| \leq k$ such that each component of $G[V - X]$ contains at most $|Q - X|/2$ vertices from Q .

Corollary 1. Let $G(V, E)$ be a graph on n vertices. Then if $\text{tw}(G) < k$, then there exists a subset X of V with $|X| \leq k$ such that each connected component of $G[V - X]$ contains at most $(n - |X|)/2$ vertices.

Proof. Let $Q = V$ in Lemma 6. $\square$

Lemma 7. Let $G(V, E)$ be a graph with n vertices. If for each subset X of V with $n/4 \leq |X| \leq n/2$, $|N(X)| \geq k$, then $\text{tw}(G) \geq k - 1$.

Proof. Suppose $\text{tw}(G) < k - 1$. Note that $k \leq \lceil n/2 \rceil$ since for a subset Y with $|Y| = \lfloor n/2 \rfloor$, $|N(Y)| \leq \lceil n/2 \rceil$. Let X be the separator with $|X| \leq k - 1$, guaranteed by Corollary 1. Let $Y_1, Y_2, \dots, Y_h$ denote the connected components of $G[V - X]$. Let $I \subseteq \{1, 2, \dots, h\}$ be such that the cardinality of $Y = \bigcup_{i \in I} Y_i$ is minimum subject to the condition that $|Y| \geq n/4$. (Such a set Y exists since $|V - X| > n/2$.) We claim that $|Y| < n/2$: If I is a singleton set, this is clearly true, since $|Y_i| < n/2$ for $1 \leq i \leq h$, by Corollary 1. Otherwise, for any $i \in I$, $|Y - Y_i| < n/4$ and $|Y_i| < n/4$, by the assumption that $|Y|$ is minimal. Thus, again we have $|Y| \leq n/2$. Now $|N(Y)| \geq k$ by assumption, but since $N(Y) \subseteq X$, this leads to a contradiction. We infer that treewidth of G is at least $k - 1$. $\square$

Now, we are in a position to present our lower bound for the treewidth of Hamming graphs.

Theorem 3. $\text{tw}(K_q^d) \geq c_1(q^d/\sqrt{d})$ for some constant $c_1 > 0$.

Proof. By Lemma 7, $\text{tw}(K_q^d) \geq \min b_v(m, K_q^d) - 1$ over integers m in the range $q^d/4 \leq m \leq q^d/2$. So it is sufficient to give a lower bound for $\min b_v(m, K_q^d)$, over $m \in [q^d/4, q^d/2]$.

In [6], Harper studies the vertex isoperimetric problem on K_q^d . He showed¹ that if

$$m = q^d \sum_{i=0}^r \binom{d}{i} t^{d-i} (1-t)^i \quad \text{for some } t, 0 < t < 1 \quad (3)$$

then

$$b_v(m, K_q^d) \geq q^d \min_t \left\{ \binom{d}{r} t^{d-r+1} (1-t)^{r+1} \right\}. \quad (4)$$

The binomial expansion of $(1-p+p)^d$ denotes a binomial distribution with mean dp and variance $dp(1-p)$. Let E denote the event that occurs with probability p . If X is a random variable that counts the number of occurrences of the event E , then from Chebyshev's inequality, we see that

$$\Pr[|X - dp| \geq \sqrt{4dp(1-p)}] \leq \frac{1}{4}.$$

This implies that

$$\sum_{i=dp-\sqrt{4dp(1-p)}}^{dp+\sqrt{4dp(1-p)}} \binom{d}{i} (1-p)^{d-i} p^i \geq \frac{3}{4}. \quad (5)$$

Suppose $m = f \cdot q^d$, where $\frac{1}{4} \leq f \leq \frac{1}{2}$. Now, if m is to be represented using some parameters t and r (where $0 < t < 1$ and $0 \leq r \leq d$) as in the RHS of Eq. (3), i.e.,

$$m = f \cdot q^d = q^d \sum_{i=0}^r \binom{d}{i} t^{d-i} (1-t)^i$$

then, since $\frac{1}{4} \leq f \leq \frac{1}{2}$, in view of inequality (5), we get (substituting $p = (1-t)$),

$$d(1-t) - \sqrt{4d(1-t)t} < r < d(1-t) + \sqrt{4d(1-t)t}.$$

Now, using Stirling's approximation, it can be shown that for all r in the above range, $q^d \cdot \binom{d}{r+1} t^{d-r+1} (1-t)^{r+1} \geq c_1(q^d/\sqrt{d})$ where $c_1 > 0$ is a constant. Hence, in view of inequality (4), $b_v(m, K_q^d) \geq c_1(q^d/\sqrt{d})$, for any $m \in [q^d/4, q^d/2]$ and the required result follows from Lemma 7. $\square$

Remark. In the special case of $q = 2$, namely when K_q^d happens to be the hypercube, then the constants c_1 in Theorem 3 can be improved by making use of Harper's original result, described in Lemma 4. In fact we can show that for hypercube, $c_1 > 0.48$, for large d .

Theorem 4. $\text{pw}(K_q^d) \leq c_2(q^d/\sqrt{d})$ for some constant c_2 .

Proof. We show that a path decomposition with width less than $c_2(q^d/\sqrt{d})$ (for some constant c_2) exists for K_q^d . Let us assume for convenience, that q is even. (The case when q is odd can be handled similarly.)

¹ This statement is not explicitly stated in [6]. But putting together the formulas given on pp. 289 and 298, the statement we have given above can be easily inferred.

We describe below a function “ g ” which maps the vertices of K_q^d to the vertices of H_d . Let f be a function from $\{0, 1, 2, \dots, q-1\}$ to $\{0, 1\}$ defined as follows:

$$\begin{aligned} f(i) &= 0 & \text{if } 0 \leq i \leq \frac{q}{2} - 1 \\ &= 1 & \text{if } \frac{q}{2} \leq i \leq q-1. \end{aligned}$$

Suppose $(a_1, a_2, \dots, a_d)$ be the d -vector which corresponds to a vertex x of K_q^d . (Note that $a_i \in \{0, 1, \dots, q-1\}$.) Then the function g will map x to the vertex y of H_d , which corresponds to the $\{0, 1\}$ -vector $(f(a_1), f(a_2), \dots, f(a_d))$.

Note that g maps exactly $(q/2)^d$ vertices of K_q^d to a given vertex of H_d . Also, it is easy to convince oneself that there exists an edge between two vertices x and y of H_d , if and only if there exists an edge between the subsets of vertices $g^{-1}(x)$ and $g^{-1}(y)$ in K_q^d . ($g^{-1}(x)$ stands for the subset of vertices of K_q^d which are mapped to x by g . We say that there exists an edge between two subsets S_1 and S_2 , if there exists a vertex $a \in S_1$ and a vertex $b \in S_2$ such that (a, b) is an edge.)

Now we can construct a path decomposition for K_q^d , using the path decomposition of H_d constructed in Section 3. We just have to replace P_i with P'_i defined as follows:

$$P'_i = \bigcup_{x \in P_i} g^{-1}(x).$$

It is not difficult to verify that $(P'_1, P'_2, \dots, P'_{2^d})$ is indeed a path decomposition for K_q^d .

Finally, the width of this path decomposition is

$$\max_i |P'_i| = \max_i |P_i| \cdot \left(\frac{q}{2}\right)^d = \text{pw}(H_d) \cdot \frac{q^d}{2^d}.$$

But as mentioned in the remark in Section 2, $\text{pw}(H_d) \leq c_1 \cdot \frac{2^d}{\sqrt{d}}$. Thus, $\text{pw}(K_q^d) \leq c_1 \cdot (q^d / \sqrt{d})$. $\square$

Combining, Theorems 3 and 4, we have

Theorem 5. $\text{tw}(K_q^d) = \theta(q^d / \sqrt{d})$.

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